

# 3ART13 LED

Variant: RELEASED

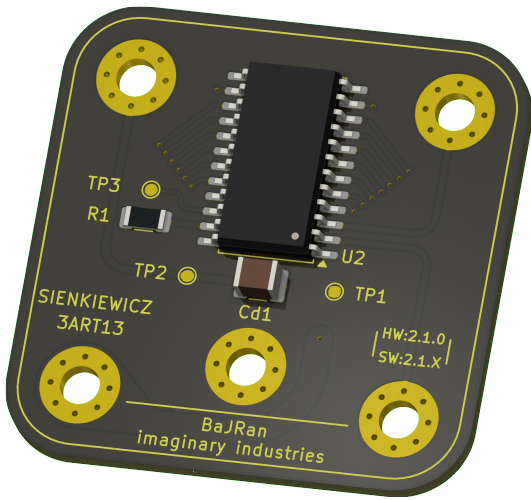
2026-05-21  
Rev 2.1.1

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## TOP VIEW



## BOTTOM VIEW



## NOTES

- Comment
- Not fitted components are marked as **X**
- DRAFT** - Very early stage of schematic, ignore details.
- PRELIMINARY** - Close to final schematic.
- CHECKED** - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED** - A board with this schematic has been sent to production.

Date: 21-May-2026

## DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

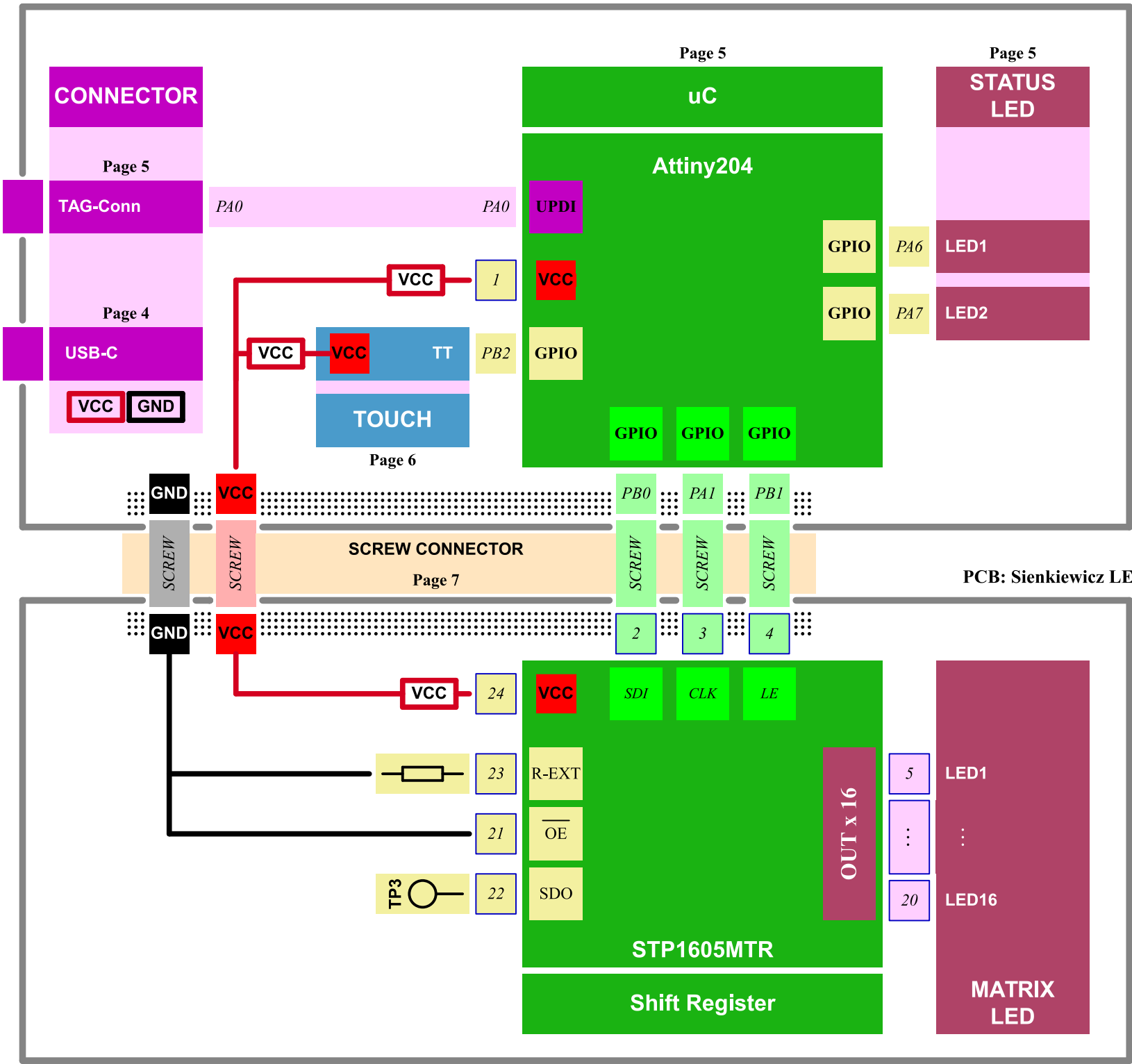
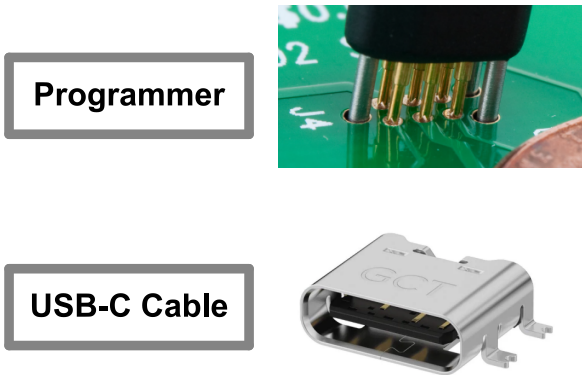
LAYOUT NOTE:

Example text for critical layout guidelines.

|  |                  |                                         |                              |                      |
|--|------------------|-----------------------------------------|------------------------------|----------------------|
|  | Comments:        | Company:<br>BaJRan imaginary industries | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|  | Sheet Title:     | Board Name:<br>3ART13 LED               | Project Name:<br>Sienkiewicz |                      |
|  | Sheet Path:<br>/ | File Name:<br>Sienkiewicz_LED.kicad_sch | Designer:<br>Pawel Baran     | Date:<br>2025-01-12  |
|  |                  |                                         | Revision:<br>2.1.1           |                      |
|  |                  |                                         | Size:<br>A3                  | Sheet:<br>1 of 8     |

[2] Block Diagram

PCB: Sienkiewicz TOUCH

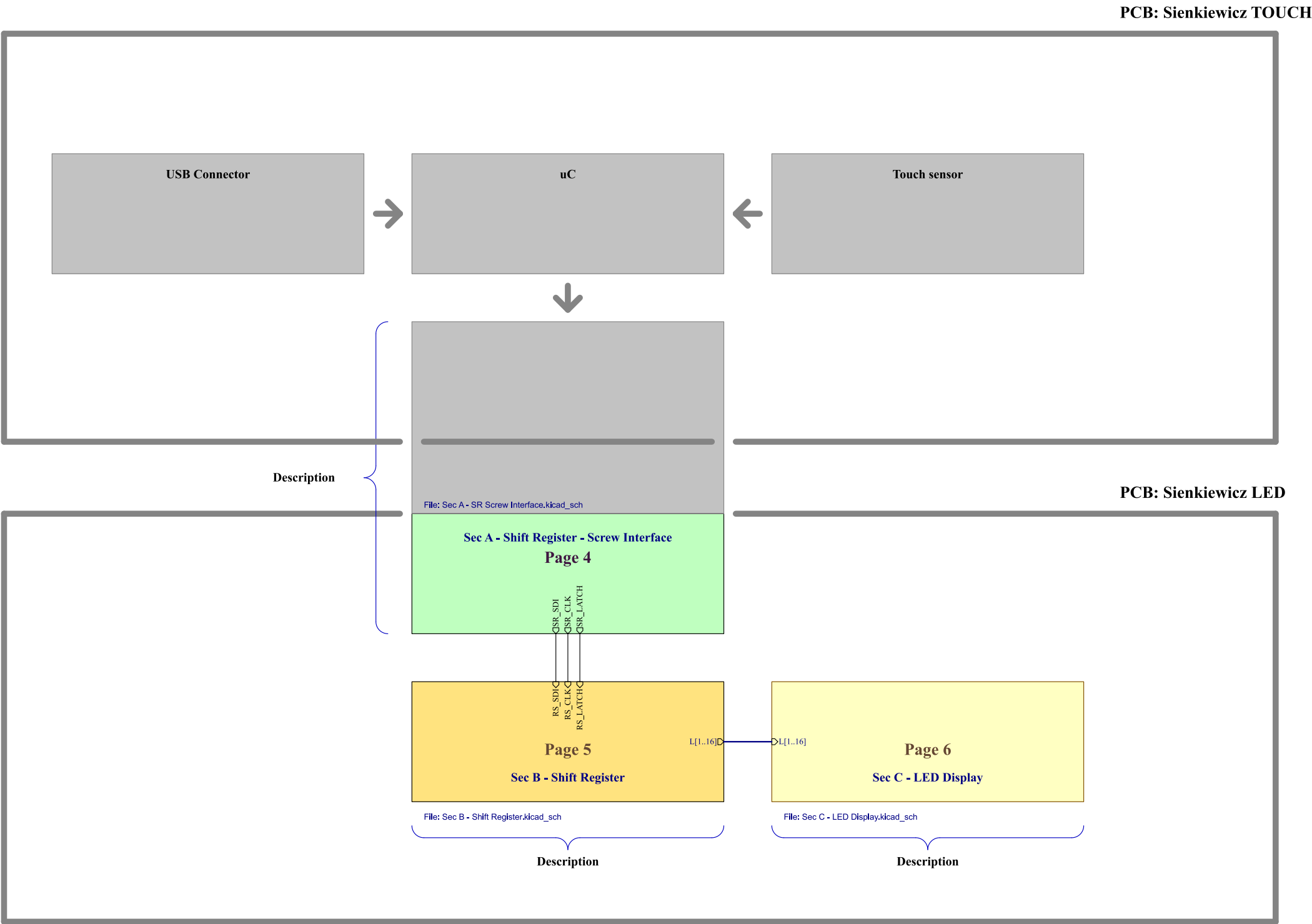


Target specifications:

|               |       |
|---------------|-------|
| Input voltage | 5V    |
| Current       | 200mA |
| Spec 3        | ----  |
| Spec 4        | ----  |

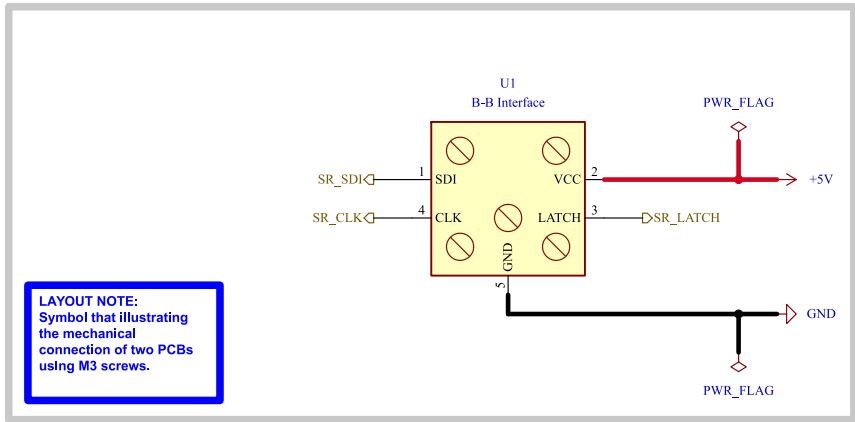
|                                |                                                                                                                               |                                       |                              |                      |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------|----------------------|
| Comments:                      | Company:<br>BaJRan  imaginary industries |                                       | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|                                | Board Name:<br>3ART13 LED                                                                                                     |                                       | Project Name:<br>Sienkiewicz |                      |
|                                | Sheet Title:<br>Block Diagram                                                                                                 | File Name:<br>Block Diagram.kicad_sch | Designer:<br>Pawel Baran     | Date:<br>2025-01-12  |
| Sheet Path:<br>/Block Diagram/ |                                                                                                                               | Reviewer:                             | Size:<br>A3                  | Revision:<br>2.1.1   |
|                                |                                                                                                                               |                                       | Sheet:<br>2 of 8             |                      |

# [3] Project Architecture



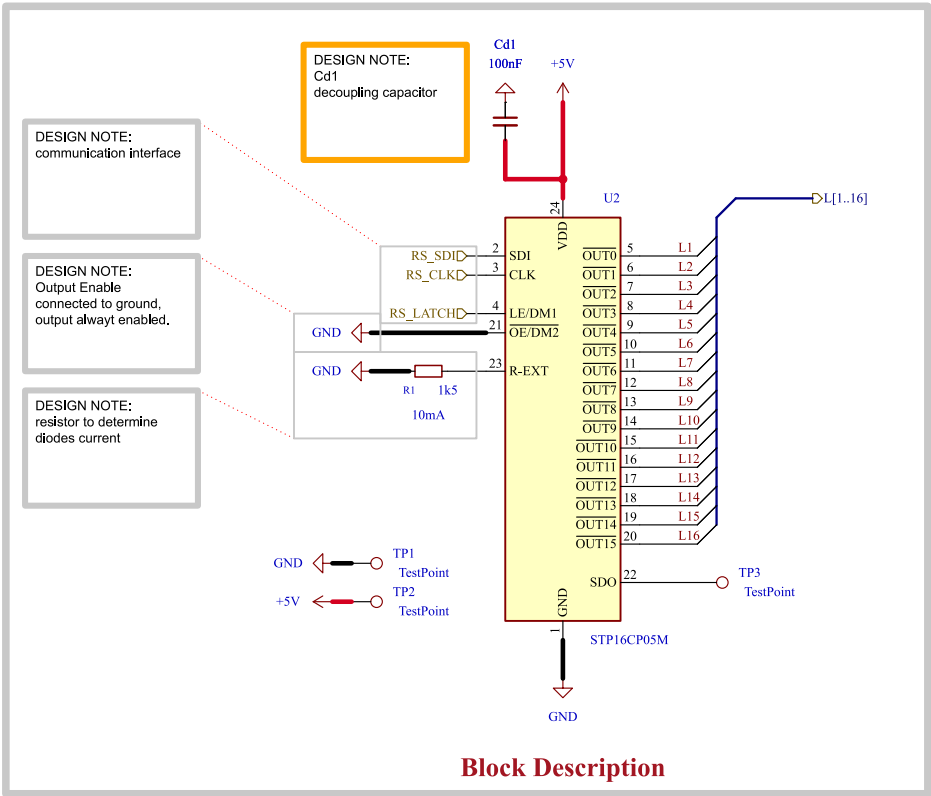
|                                       |                                                                                                                               |                                              |                              |                      |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------|----------------------|
| Comments:                             | Company:<br>BaJRan  imaginary industries |                                              | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|                                       | Board Name:<br>3ART13 LED                                                                                                     |                                              | Project Name:<br>Sienkiewicz |                      |
|                                       | Sheet Title:<br>Project Architecture                                                                                          | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Pawel Baran     | Date:<br>2025-01-12  |
| Sheet Path:<br>/Project Architecture/ |                                                                                                                               |                                              | Reviewer:                    | Size:<br>A3          |
|                                       |                                                                                                                               |                                              | Revision:                    | Sheet:<br>3 of 8     |

# [4] Shift register - Screw Interface



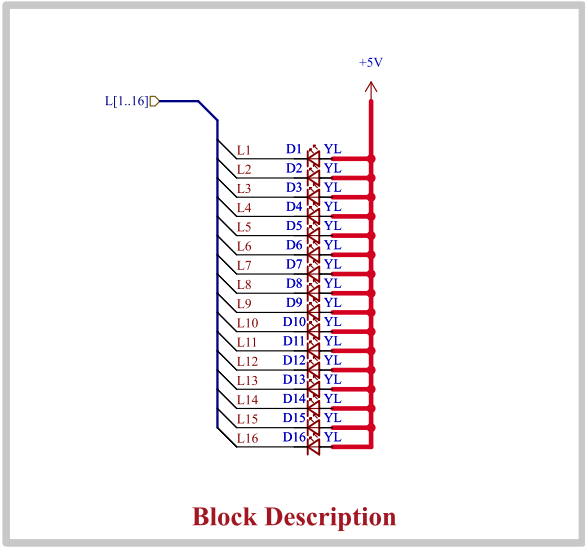
|  |                                                                                       |                                                           |                                 |                                     |                             |
|--|---------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------|-------------------------------------|-----------------------------|
|  | <b>Comments:</b><br>AN5346<br>STM32G474 Datasheet p.81<br>J. Pieper ADC investigation | <b>Company:</b><br>BaJRan imgaginary industries           |                                 | <b>Variant:</b><br>RELEASED         | <b>Git Hash:</b><br>965b332 |
|  |                                                                                       | <b>Board Name:</b><br>3ART13 LED                          |                                 | <b>Project Name:</b><br>Sienkiewicz |                             |
|  | <b>Sheet Title:</b><br>Shift register - Screw Interface                               | <b>File Name:</b><br>Sec A - SR Screw Interface.kicad_sch | <b>Designer:</b><br>Pawel Baran | <b>Date:</b><br>2025-01-12          | <b>Revision:</b><br>2.1.1   |
|  | <b>Sheet Path:</b><br>/Project Architecture/Sec A - Shift Register - Screw Interface/ |                                                           | <b>Reviewer:</b>                | <b>Size:</b><br>A4                  | <b>Sheet:</b><br>4 of 8     |

# [5] Shift Register



|  |                                                              |                                                |                          |                              |                      |
|--|--------------------------------------------------------------|------------------------------------------------|--------------------------|------------------------------|----------------------|
|  | Comments:                                                    | Company:<br>BaJRan imgaginary industries       |                          | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|  |                                                              | Board Name:<br>3ART13 LED                      |                          | Project Name:<br>Sienkiewicz |                      |
|  | Sheet Title:<br>Shift Register                               | File Name:<br>Sec B - Shift Register.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2025-01-12          | Revision:<br>2.1.1   |
|  | Sheet Path:<br>/Project Architecture/Sec B - Shift Register/ |                                                | Reviewer:                | Size:<br>A4                  | Sheet:<br>5 of 8     |

# [6] LED Display



Block Description

|  |                                                           |                                             |                          |                              |                      |
|--|-----------------------------------------------------------|---------------------------------------------|--------------------------|------------------------------|----------------------|
|  | Comments:                                                 | Company:<br>BaJRan imgaginary industries    |                          | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|  |                                                           | Board Name:<br>3ART13 LED                   |                          | Project Name:<br>Sienkiewicz |                      |
|  | Sheet Title:<br>LED Display                               | File Name:<br>Sec C - LED Display.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2026-04-30          | Revision:<br>2.1.1   |
|  | Sheet Path:<br>/Project Architecture/Sec C - LED Display/ |                                             | Reviewer:                | Size:<br>A4                  | Sheet:<br>6 of 8     |

# [7] Requirements

|                |                                                                                                                                                                                                                                                           |      |                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Project        | Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering. | MEET | The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector. |
| MAIN OBJECTIVE |                                                                                                                                                                                                                                                           |      |                                                                                                                                             |

|    |                                                                                            |      |                                                                                                                              |
|----|--------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------|
| 1  | The project shall contain very small integrated circuits.<br>Pitch less than 1mm           | MEET | IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95<br>IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27 |
| 2  | The project shall contain 0805 SMT passives components.                                    | MEET | Device contain multiple 0805 SMT elements.                                                                                   |
| 3  | The project shall contain multi-pin integrated circuits to verify soldering precision.     | MEET | IC: Shift register : STP16CP05MTR : SO-24 : 24 pins<br>IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins                      |
| 4  | The project shall contain an UPDI programming interface for firmware upload and debugging. | MEET | uC Attiny204 with UPDI programming interface                                                                                 |
| 5  | Possibility to program using Microchip enviroment and pickit-4 programmer.                 | MEET | It is possible to program the device using pickit-4 and MPLAB software.                                                      |
| 6  | Shall contain at least two status led.                                                     | MEET | Status LED'S : LED1, LED2.                                                                                                   |
| 7  | Shall be able to interact with user.                                                       | MEET | The user can change the light effect using a capacitive touch sensor.                                                        |
| 8  | Shall be easy to power up.                                                                 | MEET | Powered by USB-C connector with general mode                                                                                 |
| 9  |                                                                                            |      |                                                                                                                              |
| 10 |                                                                                            |      |                                                                                                                              |
| 11 |                                                                                            |      |                                                                                                                              |
| 12 |                                                                                            |      |                                                                                                                              |

|  |  |                                     |                                            |                          |                              |                      |
|--|--|-------------------------------------|--------------------------------------------|--------------------------|------------------------------|----------------------|
|  |  | Comments:                           | Company:<br>BaJRan imgaginary industries   |                          | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|  |  |                                     | Board Name:<br>3ART13 LED                  |                          | Project Name:<br>Sienkiewicz |                      |
|  |  | Sheet Title:<br>Requirements        | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2025-01-12          | Revision:<br>2.1.1   |
|  |  | Sheet Path:<br>/Power - Sequencing/ |                                            | Reviewer:                | Size:<br>A4                  | Sheet:<br>7 of 8     |

# [8] Revision History

## Version 2.1.0 - 2026-05-02

- Added
- First version of PCB and Schematic

## Version 2.1.1 - 2026-05-21

- Changed
- B-B connector maping changes restored

- Added
- requirements section added for documentation

|  |                                   |                                          |                          |                              |                      |
|--|-----------------------------------|------------------------------------------|--------------------------|------------------------------|----------------------|
|  | Comments:                         | Company:<br>BaJRan imgaginary industries |                          | Variant:<br>RELEASED         | Git Hash:<br>965b332 |
|  |                                   | Board Name:<br>3ART13 LED                |                          | Project Name:<br>Sienkiewicz |                      |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Pawel Baran | Date:<br>2025-01-12          | Revision:<br>2.1.1   |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:                | Size:<br>A4                  | Sheet:<br>8 of 8     |